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Using ChatGPT-4o, I simulated 500 runs of the Holt-Laury lottery on both older and younger American adults. I found that Chat was able to generate answers that, in the aggregate, partially reproduced the finding of Albert and Duffy’s real-life experiments measuring risk tolerance between younger and older adults. The ability of ChatGPT to potentially simulate behavior among different demographic lines has important implications for future economic research.

In this paper, I wanted to test the ability of LLMs to predict real human behavior given the improvements in LLMs since Horton published his paper two years ago. One of the advantages of homo silicus that Professor Horton mentions in his paper is that LLMs can be easily asked to roleplay as various demographic subgroups. In his paper, he asked the model to adapt various economic ideologies, such as libertarian and socialist, before giving them a scenario. The model's responses varied depending on their given ideology. This made me interested in what other demographic cross section LLMs could model. I wanted to examine how LLMs simulate the behavior of people of different age groups and see if they mapped to reality. On the question of short-term financial risk tolerance, there are several studies that support the idea that older Americans have a higher aversion to risk than younger Americans. One study that I looked at was called "Differences in Risk Aversion Between Young and Older Adults" by Albert and Duffy [2]. This 2012 study tested, among other things, the short-term risk tolerance of an older and

The number of times a respondent chooses A can serve as an indicator of their risk tolerance. In Holt and Laury's paper, in which they use a lottery payout proportional to the one in figure 1, they translate the number of safe choices A in terms of the respondent's risk tolerance (see figure 2). In this experiment, the authors modeled people's risk tolerances as being consistent. However, as the authors note, this is a simplified model because empirically tolerance for risk decreases as the monetary values at stake increase. However, bucketing people into these categories based on their responses to this experiment is a quick and easy way to measure their risk tolerance.

Holt-Laury Paired Lottery, Risk Aversion Task

Choice	Option A	Option B	Expected payoff Difference (choose A)
1	1/10 of \$20.00, 9/10 of \$16.00	1/10 of \$38.50, 9/10 of \$1.00	+\$11.65
2	2/10 of \$20.00, 8/10 of \$16.00	2/10 of \$38.50, 8/10 of \$1.00	+\$8.30
3	3/10 of \$20.00, 7/10 of \$16.00	3/10 of \$38.50, 7/10 of \$1.00	+\$4.95
4	4/10 of \$20.00, 6/10 of \$16.00	4/10 of \$38.50, 6/10 of \$1.00	+\$1.60
5	5/10 of \$20.00, 5/10 of \$16.00	5/10 of \$38.50, 5/10 of \$1.00	-\$1.75
6	6/10 of \$20.00, 4/10 of \$16.00	6/10 of \$38.50, 4/10 of \$1.00	-\$5.10
7	7/10 of \$20.00, 3/10 of \$16.00	7/10 of \$38.50, 3/10 of \$1.00	-\$8.45
8	8/10 of \$20.00, 2/10 of \$16.00	8/10 of \$38.50, 2/10 of \$1.00	-\$11.80
9	9/10 of \$20.00, 1/10 of \$16.00	9/10 of \$38.50, 1/10 of \$1.00	-\$15.15
10	10/10 of \$20.00, 0/10 of \$16.00	10/10 of \$38.50, 0/10 of \$1.00	-\$18.50

Figure 1: Holt-Laury paired lottery payouts. [2]

Number of safe choices	Range of relative risk aversion for $U(x) = x^{1-r}/(1-r)$	Risk preference classification	Proportion of choices		
			Low real ^a	20x hypothetical	20x real
0-1	$r < -0.95$	highly risk loving	0.01	0.03	0.01
2	$-0.95 < r < -0.49$	very risk loving	0.01	0.04	0.01
3	$-0.49 < r < -0.15$	risk loving	0.06	0.08	0.04
4	$-0.15 < r < 0.15$	risk neutral	0.26	0.29	0.13
5	$0.15 < r < 0.41$	slightly risk averse	0.26	0.16	0.19
6	$0.41 < r < 0.68$	risk averse	0.23	0.25	0.23
7	$0.68 < r < 0.97$	very risk averse	0.13	0.09	0.22
8	$0.97 < r < 1.37$	highly risk averse	0.03	0.03	0.11
9-10	$1.37 < r$	stay in bed	0.01	0.03	0.06

^a Average over first and second decisions.

Figure 2: Classifying risk tolerance based on lottery responses. [3]

3 Modifications

In the Albert and Duffy experiment, they created two groups, one consisting of younger adults and another consisting of older adults. They then instructed each subject to indicate their preference for A and B for each round. Notably, the participants in this experiment were told they would get real money based on the outcome of the lottery they selected, so their responses reflect revealed preferences. See figure 3 for a visualization of their results.

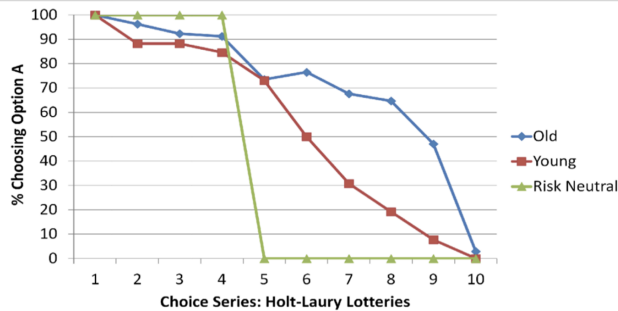


Figure 3: Results from Albert and Duffy study. [2]

As shown in Figure 3, the members of the older group chose option A at a higher rate than the younger group, meaning that they were more risk-averse. I wanted to see if I could recreate these results using LLMs primed with different age profiles.

In order to recreate this experiment, I needed to set up the two age groups. In the paper, the younger group had mean age 31.9 with standard deviation 4.7, while the older group had mean age 71.2 with standard deviation 8.4. Because the authors did not provide more detail than this, I modeled these two groups as normal random variables with the same sample means and standard deviations to get a good approximation of the two groups. I drew from these normal distributions to get ages for the experiment and then rounded them to the nearest whole number. To flesh out the personas of my respondents, I randomly chose between the respondent being a man or woman with a 50/50 probability. Inspired by the paper “Using Large Language Models to Simulate Multiple Humans and Replicate Human Subject Studies,” I randomly selected surnames from the 2010 Census Data to simulate a diverse array of American respondents [1] [6]. The opening to my prompts was as follows: “Imagine a typical {age} year-old American {man/woman} named {Mr./Ms.} {name}. Consider the risk preferences and decision-making processes of a person with these characteristics.”

I instructed ChatGPT to indicate which choice they would switch from lottery A to lottery B with a single number representing the last choice for which they would pick A (they were also allowed to choose B for all rounds by responding with 0). I decided on this approach because it forced the responses to be rational. A rational actor would not switch from A to B back to A again (because choice A becomes worse on an expected value basis with each successive round). Although some respondents in the real data engaged in illogical behavior by switching back to A, it was not as pervasive as it was with ChatGPT when I used a String-based response format. This prompting strategy was taken from the paper “Decision-Making Behavior Evaluation Framework for LLMs under Uncertain Context,” in which the authors also prompted LLMs to participate in the Holt-Laury lottery [5].

As a baseline, I decided to prompt ChatGPT without any age-based information (opening: “Imagine a typical American pronoun named name”). I asked ChatGPT this question 50 times, asking it to assume 50 different profiles. Each response was generated through a separate API call with a newly sampled respondent profile. The simulation was designed so that model responses would be independent across respondents, although I did not separately log session-level metadata to verify this assumption.

What I found was promising. ChatGPT-4o did not adhere to the EV correct behavior but instead displayed a more human-like risk aversion. On average, it chose the safe option 5.36 times. To reiterate, an EV-based approach would have chosen the safe option 4 times.

Next, I tested how ChatGPT would behave when prompted with age-specific respondent profiles.

Just as with the Albert and Duffy’s experiment, in the LLM simulations, the older group displayed greater risk aversion than the younger group. On average, the simulated younger group chose the safer option 5.578 times, while the older group chose the safer option 6.338 times. Table 1 gives a classification of the risk tolerance of virtual respondents based on Holt and Laury’s framework. Note that much of the decrease in risk aversion among older adults is driven by “stay in bed” respondents that chose A for 9 or 10 rounds.

The older group selected the safer option 0.76 more times on average than the younger group. A Mann-Whitney U test (which

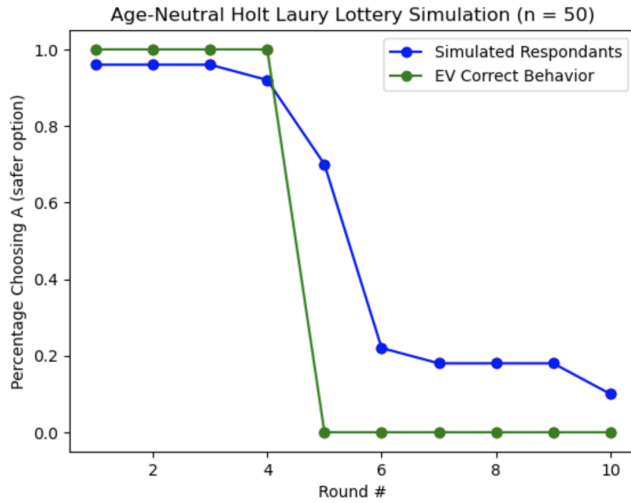


Figure 4: Running the experiment without specifying age.

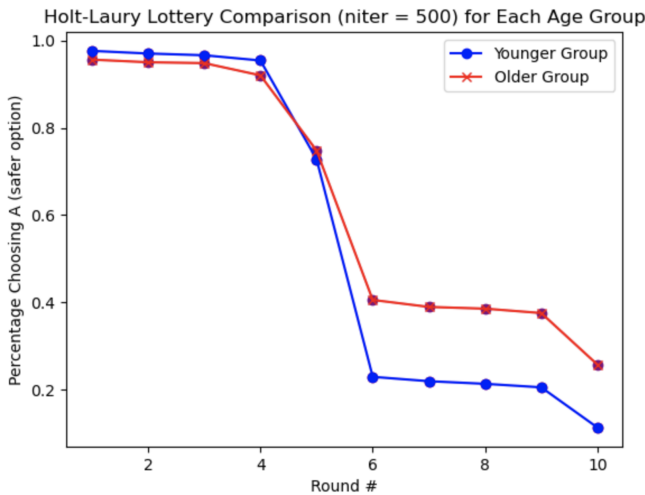


Figure 5: Running the experiment with age specified.

returned a $p = 2.638e-05$) indicated that the simulated older and younger response distributions differed significantly. This provides evidence that ChatGPT-4o's simulated economic choices were sensitive to age information in the prompt.

In the Albert and Duffy study, the 26 younger respondents surveyed chose option A 5.5 times on average, while the 34 older respondents chose option A 7.0 times on average. Using mean number of times option A selected as a metric, the absolute errors of Chat's responses were $(5.578-5.5) = 0.078$ for the younger group, and $(6.338-7.0) = 0.67$ for the older group. Specifying the age in the prompt led to more naturalistic results. For the prompt in which age was not specified, the agent chose A an average of 5.36 times, which equates to greater errors for both groups (0.14 and 1.64 respectively).

Table 1: Risk Category Distribution by Age Group (simulated)

Risk Category	Older (%)	Younger (%)
Highly risk loving	5.0	3.0
Very risk loving	0.2	0.4
Risk loving	2.8	1.2
Risk neutral	17.2	22.6
Slightly risk averse	34.2	49.8
Risk averse	1.6	1.0
Very risk averse	0.4	0.6
Highly risk averse	1.0	0.8
Stay in bed	37.6	20.6

4 Conclusion

Subjecting ChatGPT to the Holt-Laury experiment and having it roleplay as Americans of different age groups produced results fairly similar to the empirical study conducted by Albert and Duffy. ChatGPT-4o seemed to recognize that older Americans tend to display lower risk tolerance than younger Americans and approximated how many times Americans picked A in the aggregate. This finding suggests that ChatGPT-4o may be useful as a low-cost pilot tool for behavioral-economic simulations, though not a substitute for human-subject experiments. It also shows that having ChatGPT assume a specific age can make it behave in ways appropriate for that age. This is consistent with Horton's concept of homo silicus and suggests that LLMs may be useful for piloting academic studies or industry-relevant simulations.

Future research on this subject should consider running real-life experiments in tandem with LLM simulations. The study I based my comparison on was conducted in 2012. The present-day macroeconomic environment is quite different. Twenty dollars then is not worth the same as twenty dollars today. In another hypothetical example, different interest rates may affect how people value delayed rewards. Therefore, it would be interesting to run two contemporaneous experiments to see how close ChatGPT could get to the real-life results.

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